Lecture 3: Elliptic Curves

Mini-Course: Magma and the LMFDB

Eran Assaf

CTNT Summer School 2026

MIT

Lecture Overview

Elliptic Curves: Definition

Elliptic Curves over $\mathbb{Q}$: Mordell–Weil

L-Functions and the BSD Conjecture

Murmurations: an LMFDB-Driven Discovery

Complex Multiplication: Bridge from Lecture 2

Elliptic Curves: Definition

What is an Elliptic Curve?

An **elliptic curve** is a smooth projective curve of genus 1 with a chosen rational point $\mathcal{O}$.

What is an Elliptic Curve?

An **elliptic curve** is a smooth projective curve of genus 1 with a chosen rational point $\mathcal{O}$.

Over a field F (with $\operatorname{char} F \neq 2, 3$) it has a short **Weierstrass form:**

$$E : y^2 = x^3 + ax + b, \qquad \Delta_E = -16(4a^3 + 27b^2) \neq 0.$$

What is an Elliptic Curve?

An **elliptic curve** is a smooth projective curve of genus 1 with a chosen rational point $\mathcal{O}$.

Over a field F (with $\operatorname{char} F \neq 2, 3$) it has a short **Weierstrass form:**

$$E : y^2 = x^3 + ax + b, \qquad \Delta_E = -16(4a^3 + 27b^2) \neq 0.$$

Key invariants:

- **Discriminant** Δ_E (smoothness $\iff \Delta_E \neq 0$).
- ***j*-invariant** $j_E = 1728 \cdot \frac{4a^3}{4a^3+27b^2}$ (classifies E up to $\overline{F}$-isomorphism).
- **Conductor** N_E (when $F = \mathbb{Q}$): encodes the bad reduction primes weighted by reduction type.

What is an Elliptic Curve?

An **elliptic curve** is a smooth projective curve of genus 1 with a chosen rational point $\mathcal{O}$.

Over a field F (with $\operatorname{char} F \neq 2, 3$) it has a short **Weierstrass form:**

$$E : y^2 = x^3 + ax + b, \qquad \Delta_E = -16(4a^3 + 27b^2) \neq 0.$$

Key invariants:

- **Discriminant** Δ_E (smoothness $\iff \Delta_E \neq 0$).
- **j-invariant** $j_E = 1728 \cdot \frac{4a^3}{4a^3+27b^2}$ (classifies E up to $\overline{F}$-isomorphism).
- **Conductor** N_E (when $F = \mathbb{Q}$): encodes the bad reduction primes weighted by reduction type.

Group law. The points of E form an abelian group with $\mathcal{O}$ as identity:

$$P + Q + R = \mathcal{O}$$

whenever P, Q, R are collinear.

Elliptic Curves in Magma

Running example: $E : y^2 + y = x^3 - x$, conductor 37 (LMFDB label `37.a1`).

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // [a_1, a_2, a_3, a_4, a_6]
> E;
Elliptic Curve defined by y^2 + y = x^3 - x over Rational Field
> Conductor(E);
37
> Discriminant(E);
37
> jInvariant(E);
110592/37
```

Elliptic Curves in Magma

Running example: $E : y^2 + y = x^3 - x$, conductor 37 (LMFDB label *37.a1*).

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // [a_1, a_2, a_3, a_4, a_6]
> E;
Elliptic Curve defined by y^2 + y = x^3 - x over Rational Field
> Conductor(E);
37
> Discriminant(E);
37
> jInvariant(E);
110592/37
```

Why this curve?

- Smallest conductor among rank-1 elliptic curves over $\mathbb{Q}$.
- LMFDB label *37.a1* – we'll cross-reference frequently.

```
> E := EllipticCurve([0, 0, 1, -1, 0]);
> for p in [2, 5, 7, 37, 101] do
>     Ep := ChangeRing(E, GF(p));
>     printf "p = %3o: |E(F_p)| = %3o, a_p = %4o\n",
>            p, #Points(Ep), p+1-#Points(Ep);
> end for;
p =   2: |E(F_p)| =   5, a_p =   -2
p =   5: |E(F_p)| =   8, a_p =   -2
p =   7: |E(F_p)| =   9, a_p =   -1
p =  37: |E(F_p)| =  39, a_p =   -1   // bad reduction (37 | N_E)
p = 101: |E(F_p)| =  99, a_p =    3
```

Reduction mod p and Point Counting

```
> E := EllipticCurve([0, 0, 1, -1, 0]);
> for p in [2, 5, 7, 37, 101] do
>     Ep := ChangeRing(E, GF(p));
>     printf "p = %3o: |E(F_p)| = %3o, a_p = %4o\n",
>             p, #Points(Ep), p+1-#Points(Ep);
> end for;
p =   2: |E(F_p)| =   5, a_p =   -2
p =   5: |E(F_p)| =   8, a_p =   -2
p =   7: |E(F_p)| =   9, a_p =   -1
p =  37: |E(F_p)| =  39, a_p =   -1   // bad reduction (37 | N_E)
p = 101: |E(F_p)| =  99, a_p =    3
```

Hasse's theorem. For all p, $|a_p| \leq 2\sqrt{p}$.

Reduction mod p and Point Counting

```
> E := EllipticCurve([0, 0, 1, -1, 0]);
> for p in [2, 5, 7, 37, 101] do
>     Ep := ChangeRing(E, GF(p));
>     printf "p = %3o: |E(F_p)| = %3o, a_p = %4o\n",
>            p, #Points(Ep), p+1-#Points(Ep);
> end for;
p =   2: |E(F_p)| =   5, a_p =   -2
p =   5: |E(F_p)| =   8, a_p =   -2
p =   7: |E(F_p)| =   9, a_p =   -1
p =  37: |E(F_p)| =  39, a_p =   -1   // bad reduction (37 | N_E)
p = 101: |E(F_p)| =  99, a_p =    3
```

Hasse's theorem. For all p, $|a_p| \leq 2\sqrt{p}$.

Equivalent formulation. The Frobenius eigenvalues $\alpha, \bar{\alpha} \in \mathbb{C}$ satisfy $\alpha\bar{\alpha} = p$ and $\alpha + \bar{\alpha} = a_p$, hence $|\alpha| = \sqrt{p}$ (Riemann hypothesis for $E/\mathbb{F}_p$).

Reduction mod p and Point Counting

```
> E := EllipticCurve([0, 0, 1, -1, 0]);
> for p in [2, 5, 7, 37, 101] do
>     Ep := ChangeRing(E, GF(p));
>     printf "p = %3o: |E(F_p)| = %3o, a_p = %4o\n",
>            p, #Points(Ep), p+1-#Points(Ep);
> end for;
p =   2: |E(F_p)| =   5, a_p =   -2
p =   5: |E(F_p)| =   8, a_p =   -2
p =   7: |E(F_p)| =   9, a_p =   -1
p =  37: |E(F_p)| =  39, a_p =   -1   // bad reduction (37 | N_E)
p = 101: |E(F_p)| =  99, a_p =    3
```

Hasse's theorem. For all p, $|a_p| \leq 2\sqrt{p}$.

Equivalent formulation. The Frobenius eigenvalues $\alpha, \bar{\alpha} \in \mathbb{C}$ satisfy $\alpha\bar{\alpha} = p$ and $\alpha + \bar{\alpha} = a_p$, hence $|\alpha| = \sqrt{p}$ (Riemann hypothesis for $E/\mathbb{F}_p$).

Aside. The hardness of discrete log on $E(\mathbb{F}_p)$ is the basis of elliptic-curve cryptography; for the algorithmic side see the ECC mini-course. **Here we turn to $E(\mathbb{Q})$ – what are the rational points?**

Elliptic Curves over $\mathbb{Q}$: Mordell–Weil

The Mordell–Weil Theorem

Theorem (Mordell 1922, Weil 1929)

For $E/\mathbb{Q}$ (or any number field), the group $E(\mathbb{Q})$ is finitely generated:

$$E(\mathbb{Q}) \simeq E(\mathbb{Q})_{\mathrm{tors}} \oplus \mathbb{Z}^{r_E},$$

*where $r_E \geq 0$ is the **rank** of E and $E(\mathbb{Q})_{\mathrm{tors}}$ is the finite **torsion subgroup**.*

The Mordell–Weil Theorem

Theorem (Mordell 1922, Weil 1929)

For $E/\mathbb{Q}$ (or any number field), the group $E(\mathbb{Q})$ is finitely generated:

$$E(\mathbb{Q}) \simeq E(\mathbb{Q})_{\mathrm{tors}} \oplus \mathbb{Z}^{r_E},$$

*where $r_E \geq 0$ is the **rank** of E and $E(\mathbb{Q})_{\mathrm{tors}}$ is the finite **torsion subgroup**.*

Two natural questions.

- What are the possible torsion subgroups?
- What is the rank?

The Mordell–Weil Theorem

Theorem (Mordell 1922, Weil 1929)

For $E/\mathbb{Q}$ (or any number field), the group $E(\mathbb{Q})$ is finitely generated:

$$E(\mathbb{Q}) \simeq E(\mathbb{Q})_{\mathrm{tors}} \oplus \mathbb{Z}^{r_E},$$

*where $r_E \geq 0$ is the **rank** of E and $E(\mathbb{Q})_{\mathrm{tors}}$ is the finite **torsion subgroup**.*

Two natural questions.

- What are the possible torsion subgroups?
- What is the rank?

Status. Torsion is fully classified (Mazur 1977). Rank is open – no known algorithm *guaranteed* to compute it.

Theorem (Mazur, 1977)

For $E/\mathbb{Q}$, $E(\mathbb{Q})_{\mathrm{tors}}$ is one of 15 possible groups:

- $\mathbb{Z}/n\mathbb{Z}$ *for* $n \in \{1, 2, \ldots, 10, 12\}$;
- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n\mathbb{Z}$ *for* $n \in \{1, 2, 3, 4\}$.

Theorem (Mazur, 1977)

For $E/\mathbb{Q}$, $E(\mathbb{Q})_{\mathrm{tors}}$ is one of 15 possible groups:

- $\mathbb{Z}/n\mathbb{Z}$ *for* $n \in \{1, 2, \ldots, 10, 12\}$;
- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n\mathbb{Z}$ *for* $n \in \{1, 2, 3, 4\}$.

Strikingly missing. $\mathbb{Z}/11\mathbb{Z}$, $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/14\mathbb{Z}$, ...– these never occur over $\mathbb{Q}$.

Torsion Subgroups: Mazur's Theorem

Theorem (Mazur, 1977)

For $E/\mathbb{Q}$, $E(\mathbb{Q})_{\text{tors}}$ is one of 15 possible groups:

- $\mathbb{Z}/n\mathbb{Z}$ *for* $n \in \{1, 2, \ldots, 10, 12\}$;
- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n\mathbb{Z}$ *for* $n \in \{1, 2, 3, 4\}$.

Strikingly missing. $\mathbb{Z}/11\mathbb{Z}$, $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/14\mathbb{Z}$, ...– these never occur over $\mathbb{Q}$.

Geometric meaning. The modular curve $X_1(N)$ parametrizes pairs (E, P) with P of order N. Mazur's theorem $\iff X_1(N)(\mathbb{Q})$ has only cusps for $N \in \{11, 13, 14, 15, 16, 18, 20, 21, 24, \ldots\}$ – a deep fact about modular curves whose arithmetic we'll revisit in Lecture 4.

Computing Torsion in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> T, phi := TorsionSubgroup(E);
> T;
Abelian Group of order 1
> // Trivial torsion: |E(Q)_tors| = 1
```

Computing Torsion in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> T, phi := TorsionSubgroup(E);
> T;
Abelian Group of order 1
> // Trivial torsion: |E(Q)_tors| = 1
```

```
> // A curve with non-trivial torsion: 11.a3, y^2 + y = x^3 - x^2
> E2 := EllipticCurve([0, -1, 1, 0, 0]);
> T2, phi2 := TorsionSubgroup(E2);
> T2;
Abelian Group isomorphic to Z/5
> [phi2(g) : g in T2];
[ (0 : 0 : 1), (0 : -1 : 1), (1 : -1 : 1), (1 : 0 : 1), (0 : 1 : 0) ]
```

Computing Torsion in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);     // 37.a1
> T, phi := TorsionSubgroup(E);
> T;
Abelian Group of order 1
> // Trivial torsion: |E(Q)_tors| = 1
```

```
> // A curve with non-trivial torsion: 11.a3, y^2 + y = x^3 - x^2
> E2 := EllipticCurve([0, -1, 1, 0, 0]);
> T2, phi2 := TorsionSubgroup(E2);
> T2;
Abelian Group isomorphic to Z/5
> [phi2(g) : g in T2];
[ (0 : 0 : 1), (0 : -1 : 1), (1 : -1 : 1), (1 : 0 : 1), (0 : 1 : 0) ]
```

Mazur in action. The torsion is $\mathbb{Z}/5\mathbb{Z}$, on Mazur's list. No elliptic curve over $\mathbb{Q}$ has a rational point of order $11, 13, 14, \ldots$.

Computing the Rank

Computing r_E rigorously is *hard*.

Computing the Rank

Computing r_E rigorously is *hard*.

Two approaches:

- **Descent** (typically 2-descent): provides upper and lower bounds on r_E. Often sharp; sometimes leaves a gap (Selmer rank vs. Mordell–Weil rank).
- **Analytic rank**: order of vanishing of $L(E, s)$ at $s = 1$. Conjecturally equals r_E (BSD).

Computing the Rank

Computing r_E rigorously is *hard*.

Two approaches:

- **Descent** (typically 2-descent): provides upper and lower bounds on r_E. Often sharp; sometimes leaves a gap (Selmer rank vs. Mordell–Weil rank).
- **Analytic rank**: order of vanishing of $L(E, s)$ at $s = 1$. Conjecturally equals r_E (BSD).

Conjecture (Birch–Swinnerton-Dyer, weak form)

$$\mathrm{ord}_{s=1} L(E, s) \;=\; r_E.$$

Computing the Rank

Computing r_E rigorously is *hard*.

Two approaches:

- **Descent** (typically 2-descent): provides upper and lower bounds on r_E. Often sharp; sometimes leaves a gap (Selmer rank vs. Mordell–Weil rank).
- **Analytic rank**: order of vanishing of $L(E, s)$ at $s = 1$. Conjecturally equals r_E (BSD).

Conjecture (Birch–Swinnerton-Dyer, weak form)

$$\mathrm{ord}_{s=1} L(E, s) \; = \; r_E.$$

Status. Proved when analytic rank is 0 or 1 (Gross–Zagier, Kolyvagin, 1986–1989). Open in general.

Computing the Rank

Computing r_E rigorously is *hard*.

Two approaches:

- **Descent** (typically 2-descent): provides upper and lower bounds on r_E. Often sharp; sometimes leaves a gap (Selmer rank vs. Mordell–Weil rank).
- **Analytic rank**: order of vanishing of $L(E, s)$ at $s = 1$. Conjecturally equals r_E (BSD).

Conjecture (Birch–Swinnerton-Dyer, weak form)

$$\mathrm{ord}_{s=1} L(E, s) = r_E.$$

Status. Proved when analytic rank is 0 or 1 (Gross–Zagier, Kolyvagin, 1986–1989). Open in general.

Research vista. Goldfeld conjectured that the average rank is 1/2; Bhargava–Shankar (2015) proved the average is at most ≈ 0.885. Whether ranks are bounded is wide open – the current record is rank ≥ 29 (Elkies–Klagsbrun, August 2024), breaking Elkies's 2006 record of 28.

Rank and Generators in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> // Analytic rank (uses functional equation + numerical L-values)
> AnalyticRank(E);
1 0.30600
> // Mordell-Weil generators (rigorous via 2-descent)
> rank, gens := MordellWeilShaInformation(E);
> rank;
1
> gens;
[ (0 : -1 : 1) ]
> // Heights of generators
> P := gens[1];
> CanonicalHeight(P);
0.0511114082399688...
```

Rank and Generators in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> // Analytic rank (uses functional equation + numerical L-values)
> AnalyticRank(E);
1 0.30600
> // Mordell-Weil generators (rigorous via 2-descent)
> rank, gens := MordellWeilShaInformation(E);
> rank;
1
> gens;
[ (0 : -1 : 1) ]
> // Heights of generators
> P := gens[1];
> CanonicalHeight(P);
0.0511114082399688...
```

Rank-1 generator. The point $(0, -1)$ has infinite order; every rational point on E is $n \cdot (0, -1)$ plus torsion (which is trivial here).

Rank and Generators in Magma

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> // Analytic rank (uses functional equation + numerical L-values)
> AnalyticRank(E);
1 0.30600
> // Mordell-Weil generators (rigorous via 2-descent)
> rank, gens := MordellWeilShaInformation(E);
> rank;
1
> gens;
[ (0 : -1 : 1) ]
> // Heights of generators
> P := gens[1];
> CanonicalHeight(P);
0.0511114082399688...
```

Rank-1 generator. The point $(0, -1)$ has infinite order; every rational point on E is $n \cdot (0, -1)$ plus torsion (which is trivial here).

Multiplying. $2 \cdot (0, -1) = (1, -1)$, $3 \cdot (0, -1) = (-1, 0)$, ...– the curve has infinitely many rational points, all generated by one!

L-Functions and the BSD Conjecture

The L-Function of an Elliptic Curve

For $E/\mathbb{Q}$ of conductor N, define

$$L(E,s) \;=\; \prod_{p\nmid N} \frac{1}{1-a_p p^{-s}+p^{1-2s}} \cdot \prod_{p\mid N} \frac{1}{1-a_p p^{-s}}.$$

The L-Function of an Elliptic Curve

For $E/\mathbb{Q}$ of conductor N, define

$$L(E,s) = \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \cdot \prod_{p \mid N} \frac{1}{1 - a_p p^{-s}}.$$

This converges for $\Re(s) > 3/2$. The big theorem:

Theorem (Modularity, Wiles–Taylor–Breuil–Conrad–Diamond, 2001)

$L(E,s)$ analytically continues to all of $\mathbb{C}$ and satisfies a functional equation $s \leftrightarrow 2-s$. Equivalently, $L(E,s) = L(f,s)$ for some weight-2 newform f of level N.

The L-Function of an Elliptic Curve

For $E/\mathbb{Q}$ of conductor N, define

$$L(E,s) \;=\; \prod_{p\nmid N} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \;\cdot\; \prod_{p\mid N} \frac{1}{1 - a_p p^{-s}}.$$

This converges for $\Re(s) > 3/2$. The big theorem:

Theorem (Modularity, Wiles–Taylor–Breuil–Conrad–Diamond, 2001)

$L(E,s)$ analytically continues to all of $\mathbb{C}$ and satisfies a functional equation $s \leftrightarrow 2-s$. Equivalently, $L(E,s) = L(f,s)$ for some weight-2 newform f of level N.

Consequence. We can evaluate $L(E,s)$ – and its derivatives – at $s=1$, even though $s=1$ is outside the region of convergence.

The L-Function of an Elliptic Curve

For $E/\mathbb{Q}$ of conductor N, define

$$L(E,s) \;=\; \prod_{p\nmid N} \frac{1}{1-a_p p^{-s}+p^{1-2s}} \;\cdot\; \prod_{p\mid N} \frac{1}{1-a_p p^{-s}}.$$

This converges for $\Re(s) > 3/2$. The big theorem:

Theorem (Modularity, Wiles–Taylor–Breuil–Conrad–Diamond, 2001)

$L(E,s)$ analytically continues to all of $\mathbb{C}$ and satisfies a functional equation $s \leftrightarrow 2-s$. Equivalently, $L(E,s) = L(f,s)$ for some weight-2 newform f of level N.

Consequence. We can evaluate $L(E,s)$ – and its derivatives – at $s=1$, even though $s=1$ is outside the region of convergence.

Aside. Modularity for semistable curves was Wiles's 1995 proof of Fermat's Last Theorem.

The L-Function of an Elliptic Curve

For $E/\mathbb{Q}$ of conductor N, define

$$L(E,s) = \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \cdot \prod_{p \mid N} \frac{1}{1 - a_p p^{-s}}.$$

This converges for $\Re(s) > 3/2$. The big theorem:

Theorem (Modularity, Wiles–Taylor–Breuil–Conrad–Diamond, 2001)

$L(E,s)$ analytically continues to all of $\mathbb{C}$ and satisfies a functional equation $s \leftrightarrow 2 - s$. Equivalently, $L(E,s) = L(f,s)$ for some weight-2 newform f of level N.

Consequence. We can evaluate $L(E,s)$ – and its derivatives – at $s = 1$, even though $s = 1$ is outside the region of convergence.

Aside. Modularity for semistable curves was Wiles's 1995 proof of Fermat's Last Theorem.

Bridge to Lecture 4. The newform f associated to E lives in a space of modular forms; LMFDB's modular form pages link to elliptic curves via this correspondence.

The Birch–Swinnerton-Dyer Conjecture

Conjecture (BSD, full form)

Let $E/\mathbb{Q}$ have rank r. Then

$$\lim_{s\to 1}\frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E\cdot \mathrm{Reg}_E\cdot \prod_p c_p \cdot \#Ш(E/\mathbb{Q})}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2}.$$

The Birch–Swinnerton-Dyer Conjecture

Conjecture (BSD, full form)

Let $E/\mathbb{Q}$ have rank r. Then

$$\lim_{s\to 1} \frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E \cdot \mathrm{Reg}_E \cdot \prod_p c_p \cdot \#Ш(E/\mathbb{Q})}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2}.$$

Each factor:

- Ω_E – real period (an integral on $E(\mathbb{R})$), doubled when $E(\mathbb{R})$ has two components;
- Reg_E – regulator (determinant of canonical heights of generators);
- c_p – Tamagawa numbers (local factors at bad primes);
- $\#Ш(E/\mathbb{Q})$ – order of the Tate–Shafarevich group, conjecturally finite.

Conjecture (BSD, full form)

Let $E/\mathbb{Q}$ have rank r. Then

$$\lim_{s\to 1} \frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E \cdot \mathrm{Reg}_E \cdot \prod_p c_p \cdot \#Ш(E/\mathbb{Q})}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2}.$$

Each factor:

- Ω_E – real period (an integral on $E(\mathbb{R})$), doubled when $E(\mathbb{R})$ has two components;
- Reg_E – regulator (determinant of canonical heights of generators);
- c_p – Tamagawa numbers (local factors at bad primes);
- $\#Ш(E/\mathbb{Q})$ – order of the Tate–Shafarevich group, conjecturally finite.

One of the seven Clay Millennium Problems. $1,000,000 for a proof.

Numerical BSD: Verifying $\#Ш = 1$ for `37.a1`

Strategy: compute every term except $\#Ш$, then *predict* $\#Ш$ from BSD.

```
> E := EllipticCurve([0, 0, 1, -1, 0]);    // 37.a1
> L1, _ := AnalyticRank(E);                // r = 1
> L1prime := Evaluate(LSeries(E), 1 : Derivative := 1);
> L1prime;
0.305999773834052301...
> Omega := 2*RealPeriod(E);
> Omega;
5.98691729246391925 9...
> Reg := CanonicalHeight(Generators(E)[1]);
> Tam := &*[TamagawaNumber(E, p) : p in BadPrimes(E)];
> Tors := #TorsionSubgroup(E);
> // BSD predicts: L'(E,1) = Omega * Reg * Tam * #Sha / Tors^2
> sha_predicted := L1prime * Tors^2 / (Omega * Reg * Tam);
> sha_predicted;
1.00000000000000...
```

Numerical BSD: Verifying $\#Ш = 1$ for 37.a1

Strategy: compute every term except $\#Ш$, then *predict* $\#Ш$ from BSD.

```
> E := EllipticCurve([0, 0, 1, -1, 0]);     // 37.a1
> L1, _ := AnalyticRank(E);                 // r = 1
> L1prime := Evaluate(LSeries(E), 1 : Derivative := 1);
> L1prime;
0.305999773834052301...
> Omega := 2*RealPeriod(E);
> Omega;
5.98691729246391925 9...
> Reg := CanonicalHeight(Generators(E)[1]);
> Tam := &*[TamagawaNumber(E, p) : p in BadPrimes(E)];
> Tors := #TorsionSubgroup(E);
> // BSD predicts: L'(E,1) = Omega * Reg * Tam * #Sha / Tors^2
> sha_predicted := L1prime * Tors^2 / (Omega * Reg * Tam);
> sha_predicted;
1.00000000000000...
```

BSD says $\#Ш = 1$. Computed numerically: ≈ 1.0000. This is a nontrivial verification of a Millennium Problem!

Beyond Kolyvagin: BSD as the Only Tool ($r = 3$)

For analytic rank ≥ 2, Kolyvagin's theorem does not apply: we do not even know Ш is finite. The BSD-predicted value is the *only information we have*.

Beyond Kolyvagin: BSD as the Only Tool ($r = 3$)

For analytic rank ≥ 2, Kolyvagin's theorem does not apply: we do not even know Ш is finite. The BSD-predicted value is the *only information we have*.

Smallest-conductor rank-3 curve over $\mathbb{Q}$: *5077.a1*, $E : y^2 + y = x^3 - 7x + 6$.

```
> E := EllipticCurve([0, 0, 1, -7, 6]);          // 5077.a1
> r, Lr := AnalyticRank(E : Precision := 12);     // L^(r)(1)/r!
> r;
3
> // For real-component count: Disc > 0 => 2 components
> Omega := RealPeriod(E) * (Discriminant(E) gt 0 select 2 else 1);
> Reg := Regulator(E); Tam := &*TamagawaNumbers(E);
> Tors := #TorsionSubgroup(E);
> Lr * Tors^2 / (Omega * Reg * Tam);
1.000000000...
```

Beyond Kolyvagin: BSD as the Only Tool ($r = 3$)

For analytic rank ≥ 2, Kolyvagin's theorem does not apply: we do not even know Ш is finite. The BSD-predicted value is the *only information we have*.

Smallest-conductor rank-3 curve over $\mathbb{Q}$: `5077.a1`, $E : y^2 + y = x^3 - 7x + 6$.

```
> E := EllipticCurve([0, 0, 1, -7, 6]);              // 5077.a1
> r, Lr := AnalyticRank(E : Precision := 12);      // L^(r)(1)/r!
> r;
3
> // For real-component count: Disc > 0 => 2 components
> Omega := RealPeriod(E) * (Discriminant(E) gt 0 select 2 else 1);
> Reg := Regulator(E); Tam := &*TamagawaNumbers(E);
> Tors := #TorsionSubgroup(E);
> Lr * Tors^2 / (Omega * Reg * Tam);
1.000000000...
```

What this says. BSD predicts $\#Ш(\texttt{5077.a1}) = 1$. We cannot *prove* it – but the numerical agreement to ~ 10 digits is strong evidence for both finiteness and triviality.

Beyond Kolyvagin: BSD as the Only Tool ($r = 3$)

For analytic rank ≥ 2, Kolyvagin's theorem does not apply: we do not even know Ш is finite. The BSD-predicted value is the *only information we have*.

Smallest-conductor rank-3 curve over $\mathbb{Q}$: `5077.a1`, $E : y^2 + y = x^3 - 7x + 6$.

```
> E := EllipticCurve([0, 0, 1, -7, 6]);            // 5077.a1
> r, Lr := AnalyticRank(E : Precision := 12);      // L^(r)(1)/r!
> r;
3
> // For real-component count: Disc > 0 => 2 components
> Omega := RealPeriod(E) * (Discriminant(E) gt 0 select 2 else 1);
> Reg := Regulator(E); Tam := &*TamagawaNumbers(E);
> Tors := #TorsionSubgroup(E);
> Lr * Tors^2 / (Omega * Reg * Tam);
1.000000000...
```

What this says. BSD predicts $\#Ш($`5077.a1`$) = 1$. We cannot *prove* it – but the numerical agreement to ~ 10 digits is strong evidence for both finiteness and triviality.

This is the research frontier. For rank ≥ 2, every BSD-computation in Magma is a conjectural statement. The LMFDB tabulates these analytic $\#Ш$ values; checking how often they're integer-valued is itself evidence for BSD.

Murmurations: an LMFDB-Driven Discovery

In April 2022, while running machine-learning experiments on LMFDB data, Pozdnyakov (then an undergraduate) noticed a striking pattern.

In April 2022, while running machine-learning experiments on LMFDB data, Pozdnyakov (then an undergraduate) noticed a striking pattern.

The phenomenon. Fix a conductor range $[N_1, N_2]$ and a rank r. For each prime p, average the Frobenius trace $a_p(E)$ over all $E/\mathbb{Q}$ in the range with rank r:

$$\overline{a}_r(p) := \frac{1}{\#\mathcal{E}_r} \sum_{E \in \mathcal{E}_r} a_p(E), \qquad \mathcal{E}_r = \{E : N(E) \in [N_1, N_2],\ \mathrm{rank}\, E = r\}.$$

In April 2022, while running machine-learning experiments on LMFDB data, Pozdnyakov (then an undergraduate) noticed a striking pattern.

The phenomenon. Fix a conductor range $[N_1, N_2]$ and a rank r. For each prime p, average the Frobenius trace $a_p(E)$ over all $E/\mathbb{Q}$ in the range with rank r:

$$\overline{a}_r(p) \;:=\; \frac{1}{\#\mathcal{E}_r}\sum_{E\in\mathcal{E}_r} a_p(E), \qquad \mathcal{E}_r = \{E : N(E) \in [N_1, N_2],\ \mathrm{rank}\, E = r\}.$$

As p varies, $\overline{a}_r(p)$ *oscillates* – and the shape of the oscillation depends on r. The pattern is so visible that, as Thomas Oliver put it, “it should have been noticed long ago.”

Murmurations of Elliptic Curves

In April 2022, while running machine-learning experiments on LMFDB data, Pozdnyakov (then an undergraduate) noticed a striking pattern.

The phenomenon. Fix a conductor range $[N_1, N_2]$ and a rank r. For each prime p, average the Frobenius trace $a_p(E)$ over all $E/\mathbb{Q}$ in the range with rank r:

$$\overline{a}_r(p) \;:=\; \frac{1}{\#\mathcal{E}_r}\sum_{E\in\mathcal{E}_r} a_p(E), \qquad \mathcal{E}_r = \{E : N(E) \in [N_1, N_2],\ \operatorname{rank} E = r\}.$$

As p varies, $\overline{a}_r(p)$ *oscillates* – and the shape of the oscillation depends on r. The pattern is so visible that, as Thomas Oliver put it, "it should have been noticed long ago."

Why it stayed hidden. The murmuration only appears when curves are sorted by *conductor*; sort by something else (height, discriminant), and it disappears. The LMFDB happened to be conductor-sorted – a piece of historical luck.

In April 2022, while running machine-learning experiments on LMFDB data, Pozdnyakov (then an undergraduate) noticed a striking pattern.

The phenomenon. Fix a conductor range $[N_1, N_2]$ and a rank r. For each prime p, average the Frobenius trace $a_p(E)$ over all $E/\mathbb{Q}$ in the range with rank r:

$$\overline{a}_r(p) := \frac{1}{\#\mathcal{E}_r} \sum_{E \in \mathcal{E}_r} a_p(E), \qquad \mathcal{E}_r = \{E : N(E) \in [N_1, N_2],\ \mathrm{rank}\, E = r\}.$$

As p varies, $\overline{a}_r(p)$ *oscillates* – and the shape of the oscillation depends on r. The pattern is so visible that, as Thomas Oliver put it, "it should have been noticed long ago."

Why it stayed hidden. The murmuration only appears when curves are sorted by *conductor*; sort by something else (height, discriminant), and it disappears. The LMFDB happened to be conductor-sorted – a piece of historical luck.

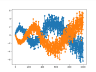
He–Lee–Oliver–Pozdnyakov, *Murmurations of Elliptic Curves*, Exp. Math. (2022, arXiv:2204.10140). Heuristic explanation: Sawin–Sutherland (2025, arXiv:2504.12295).

Murmurations in 30 Lines

```
sage: from lmfdb import db
sage: conds = (7500, 10000)
sage: query = {"conductor": {"$gte": conds[0], "$lt": conds[1]}}
sage: cols = ["rank", "ainvs"]
sage: curves_by_rank = {0: [], 1: []}
sage: for f in db.ec_curvedata.search(query, cols):
....:     if f["rank"] in curves_by_rank:
....:         curves_by_rank[f["rank"]].append(EllipticCurve(f["ainvs"]))
sage: primes = primes_first_n(1000)
sage: avg = {r: [mean([E.ap(p) for E in Es]) for p in primes] for r, Es in
    curves_by_rank.items()}
```

Murmurations in 30 Lines

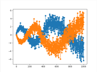
```
sage: from lmfdb import db
sage: conds = (7500, 10000)
sage: query = {"conductor": {"$gte": conds[0], "$lt": conds[1]}}
sage: cols = ["rank", "ainvs"]
sage: curves_by_rank = {0: [], 1: []}
sage: for f in db.ec_curvedata.search(query, cols):
....:     if f["rank"] in curves_by_rank:
....:         curves_by_rank[f["rank"]].append(EllipticCurve(f["ainvs"]))
sage: primes = primes_first_n(1000)
sage: avg = {r: [mean([E.ap(p) for E in Es]) for p in primes] for r, Es in
    curves_by_rank.items()}
```



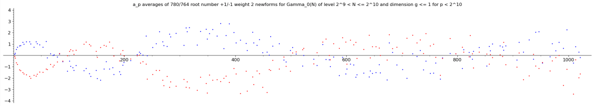
Our 7.5k–10k conductor range, ranks 0 and 1.

Murmurations in 30 Lines

```
sage: from lmfdb import db
sage: conds = (7500, 10000)
sage: query = {"conductor": {"$gte": conds[0], "$lt": conds[1]}}
sage: cols = ["rank", "ainvs"]
sage: curves_by_rank = {0: [], 1: []}
sage: for f in db.ec_curvedata.search(query, cols):
....:     if f["rank"] in curves_by_rank:
....:         curves_by_rank[f["rank"]].append(EllipticCurve(f["ainvs"]))
sage: primes = primes_first_n(1000)
sage: avg = {r: [mean([E.ap(p) for E in Es]) for p in primes] for r, Es in
    curves_by_rank.items()}
```



Our 7.5k–10k conductor range, ranks 0 and 1.



Sutherland's plot using ∼300M curves.

The pedagogical point. A modest LMFDB query + a few lines of Python and an undergraduate discovered a new phenomenon in 2022. The database *is* the laboratory.

Complex Multiplication: Bridge from Lecture 2

From BSD to CM: Closing the Loop with Lecture 2

BSD is one big-picture theme. The other is the *dictionary* between elliptic curves and number fields.

Lecture 2 left us with $H_{\mathbb{Q}(\sqrt{-23})} = \mathbb{Q}(\sqrt{-23})(\alpha)$, $\alpha^3 - \alpha - 1 = 0$, as an *abstract* presentation – just a primitive element.

From BSD to CM: Closing the Loop with Lecture 2

BSD is one big-picture theme. The other is the *dictionary* between elliptic curves and number fields.

Lecture 2 left us with $H_{\mathbb{Q}(\sqrt{-23})} = \mathbb{Q}(\sqrt{-23})(\alpha)$, $\alpha^3 - \alpha - 1 = 0$, as an *abstract* presentation – just a primitive element.

CM theory says $\alpha = j(\tau)$ for an explicit τ in the upper half-plane, and elliptic curves provide the natural construction of H_K for imaginary quadratic K.

Complex Multiplication

Generically $\mathbf{End}(E) = \mathbb{Z}$ (multiplication-by-n maps). But sometimes $\mathbf{End}(E)$ is larger.

Complex Multiplication

Generically $\mathbf{End}(E) = \mathbb{Z}$ (multiplication-by-n maps). But sometimes $\mathbf{End}(E)$ is larger.

Definition

E has **complex multiplication (CM)** if $\mathbf{End}_{\overline{\mathbb{Q}}}(E) \otimes \mathbb{Q}$ is an imaginary quadratic field K.

Complex Multiplication

Generically $\mathrm{End}(E) = \mathbb{Z}$ (multiplication-by-n maps). But sometimes $\mathrm{End}(E)$ is larger.

Definition

E has **complex multiplication (CM)** if $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \otimes \mathbb{Q}$ is an imaginary quadratic field K.

Theorem (CM theory). If $E/\mathbb{C}$ has CM by $\mathbb{Z}_K$ for K imaginary quadratic, then $j(E) \in \overline{\mathbb{Q}}$, and

$$K(j(E)) = H_K, \quad \text{the Hilbert class field of } K.$$

Complex Multiplication

Generically $\mathrm{End}(E) = \mathbb{Z}$ (multiplication-by-n maps). But sometimes $\mathrm{End}(E)$ is larger.

Definition

E has **complex multiplication (CM)** if $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \otimes \mathbb{Q}$ is an imaginary quadratic field K.

Theorem (CM theory). If $E/\mathbb{C}$ has CM by $\mathbb{Z}_K$ for K imaginary quadratic, then $j(E) \in \overline{\mathbb{Q}}$, and

$$K(j(E)) \;=\; H_K, \quad \text{the Hilbert class field of } K.$$

The dictionary, made tangible. CM points $\tau \in \mathcal{H}$ correspond to lattices $[1, \tau] \subset \mathbb{C}$ with $\mathrm{End} = \mathbb{Z}_K$, hence to ideal classes in $\mathrm{Cl}(K)$. For $K = \mathbb{Q}(\sqrt{-23})$, $|\,\mathrm{Cl}(K)| = 3$, so there are *three* CM points (mod $\mathrm{SL}_2(\mathbb{Z})$) – their j-values are the *three roots* of the Hilbert class polynomial $H_{-23}(x)$.

Generically $\mathrm{End}(E) = \mathbb{Z}$ (multiplication-by-n maps). But sometimes $\mathrm{End}(E)$ is larger.

Definition

E has **complex multiplication (CM)** if $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \otimes \mathbb{Q}$ is an imaginary quadratic field K.

Theorem (CM theory). If $E/\mathbb{C}$ has CM by $\mathbb{Z}_K$ for K imaginary quadratic, then $j(E) \in \overline{\mathbb{Q}}$, and

$$K(j(E)) \;=\; H_K, \qquad \text{the Hilbert class field of } K.$$

The dictionary, made tangible. CM points $\tau \in \mathcal{H}$ correspond to lattices $[1, \tau] \subset \mathbb{C}$ with $\mathrm{End} = \mathbb{Z}_K$, hence to ideal classes in $\mathrm{Cl}(K)$. For $K = \mathbb{Q}(\sqrt{-23})$, $|\mathrm{Cl}(K)| = 3$, so there are *three* CM points (mod $\mathrm{SL}_2(\mathbb{Z})$) – their j-values are the *three roots* of the Hilbert class polynomial $H_{-23}(x)$.

Kronecker's Jugendtraum. CM theory generates abelian extensions of imaginary quadratic K via j-values at CM points – the analog of Kronecker–Weber.

Hilbert Class Polynomials in Magma

For $K = \mathbb{Q}(\sqrt{-23})$, $\mathrm{Cl}(K) \simeq \mathbb{Z}/3\mathbb{Z}$, so $[H_K : K] = 3$ and $H_{-23}(x)$ has degree 3.

```
> H := HilbertClassPolynomial(-23);
> H;
T^3 + 3491750*T^2 - 5151296875*T + 12771880859375
> K<b> := QuadraticField(-23);
> // The splitting field of H over Q is H_K (degree 6 over Q)
> HK := ext< K | H >;
> assert IsIsomorphic(HK, ext< K | T^3 - T - 1>);
```

Hilbert Class Polynomials in Magma

For $K = \mathbb{Q}(\sqrt{-23})$, $\mathrm{Cl}(K) \simeq \mathbb{Z}/3\mathbb{Z}$, so $[H_K : K] = 3$ and $H_{-23}(x)$ has degree 3.

```
> H := HilbertClassPolynomial(-23);
> H;
T^3 + 3491750*T^2 - 5151296875*T + 12771880859375
> K<b> := QuadraticField(-23);
> // The splitting field of H over Q is H_K (degree 6 over Q)
> HK := ext< K | H >;
> assert IsIsomorphic(HK, ext< K | T^3 - T - 1>);
```

Striking. Same field, two presentations:

- $H_K = K(\alpha)$, $\alpha^3 - \alpha - 1 = 0$ (Lecture 2, tiny coefficients).
- $H_K = K(j_0)$, j_0 a root of $H_{-23}(x)$ (CM theory, huge coefficients).

Two different primitive elements for the same field; CM theory tells us j_0 has geometric meaning.

From Hilbert Class Polynomial to CM Curves

```
> // Build a CM elliptic curve with j-invariant j_0
> R<j> := PolynomialRing(Rationals());
> H := HilbertClassPolynomial(-23);
> Kj<j0> := NumberField(H);
> E := EllipticCurveFromjInvariant(j0);
> E;
Elliptic Curve defined by ... over Kj
> // E has CM by Z_K where K = Q(sqrt(-23))
> jInvariant(E) eq j0;
true
```

From Hilbert Class Polynomial to CM Curves

```
> // Build a CM elliptic curve with j-invariant j_0
> R<j> := PolynomialRing(Rationals());
> H := HilbertClassPolynomial(-23);
> Kj<j0> := NumberField(H);
> E := EllipticCurveFromjInvariant(j0);
> E;
Elliptic Curve defined by ... over Kj
> // E has CM by Z_K where K = Q(sqrt(-23))
> jInvariant(E) eq j0;
true
```

LMFDB. The page for $K = \mathbb{Q}(\sqrt{-23})$ links to the Hilbert class polynomial; the CM elliptic curves over the resulting cubic field have their own LMFDB pages.

From Hilbert Class Polynomial to CM Curves

```
> // Build a CM elliptic curve with j-invariant j_0
> R<j> := PolynomialRing(Rationals());
> H := HilbertClassPolynomial(-23);
> Kj<j0> := NumberField(H);
> E := EllipticCurveFromjInvariant(j0);
> E;
Elliptic Curve defined by ... over Kj
> // E has CM by Z_K where K = Q(sqrt(-23))
> jInvariant(E) eq j0;
true
```

LMFDB. The page for $K = \mathbb{Q}(\sqrt{-23})$ links to the Hilbert class polynomial; the CM elliptic curves over the resulting cubic field have their own LMFDB pages.

Lecture 4 preview. CM elliptic curves correspond to weight-2 newforms with CM by K – a special class of modular forms with very rigid structure. Browse them at *lmfdb.org/ModularForm/GL2/Q/holomorphic/?cm_discs=-23*.

Summary

Key Takeaways

1. Elliptic curves $E/\mathbb{Q}$ carry a finitely-generated abelian group $E(\mathbb{Q}) = \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$ (Mordell–Weil).
2. **Mazur:** torsion is one of 15 possibilities. **BSD:** rank is conjecturally the order of vanishing of $L(E, s)$ at $s = 1$.
3. Magma + LMFDB let us *numerically* verify BSD: compute $L^{(r)}(E, 1)$, periods, regulator, Tamagawa numbers, then read off $\#Ш$.
4. For $r \geq 2$ this is genuine research: Kolyvagin fails, and the BSD-predicted $\#Ш$ is the only information we have (e.g. $\#Ш = 1$ for `5077.a1`).
5. **LMFDB as laboratory:** murmurations – discovered by an undergraduate in 2022 – show new phenomena are still waiting in the database.
6. **CM theory** closes the loop with Lecture 2: $H_K = K(j(\tau))$ generates abelian extensions of imaginary quadratic K via j-invariants of CM elliptic curves.

Next lecture: modular forms, Hecke operators, and the modularity correspondence in action.